

ReCoMER: Reliability and Context Modeling for Multimodal Conversational Emotion Recognition

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Abstract

Multimodal conversational emotion recognition requires a model to decide not only how to combine text, audio and vision, but also whether historical context and an external prediction should be trusted for the current utterance. We present ReCoMER, a reliability- and context-aware framework that combines MHnoU with the independent cRBEF expert. MHnoU integrates feedback-conditioned context reassessment, Shapley-anchored bounded evidence routing and null-candidate history abstention. cRBEF constructs prior-relative audio-visual evidence and regulates it with a reliability gate. A bounded local class-wise fusion module then combines the two complete predictions while preserving an exact equal-weight fallback. On the frozen formal M3ED test, ReCoMER reaches 58.00% WF1 and improves over matched equal-weight fusion by 0.607 percentage points. Module-isolation experiments show conditional gains on MOSEI, MELD and IEMOCAP, while also exposing negative or neutral regimes. The resulting framework separates prediction, history admission, contribution estimation and expert fusion, making both its improvements and failure boundaries auditable.

Keywords: multimodal emotion recognition; emotion recognition in conversation; modality contribution; history abstention; reliability-aware fusion

1 Introduction

Multimodal Emotion Recognition (MER) in conversations aims to identify the emotion expressed in each utterance by integrating textual, acoustic, and visual cues with conversational context. Spoken words convey semantic content, while vocal characteristics and facial expressions provide complementary information about how that content is expressed. Previous utterances can further clarify ambiguous expressions and help interpret emotional changes across a conversation. Effective recognition therefore requires more than combining available features: a model must determine which modalities provide useful evidence and whether historical information supports the current judgment. These capabilities are relevant to applications such as empathetic dialogue systems, intelligent tutoring, and human–robot interaction.

However, modality contributions and the usefulness of conversational history vary across utterances. A modality may convey clear emotional evidence in one instance but provide weak, redundant, or conflicting information in another. Predictive confidence alone does not fully capture this variation, since a confident modality may contribute little beyond information already supplied by other modalities. Historical context introduces an additional difficulty: earlier utterances may resolve ambiguity, but they may also reinforce an outdated emotional state or introduce irrelevant information. Moreover, historical feedback changes current modality representations and can consequently alter their relative contributions. These challenges call for coordinated mechanisms that reassess representations after feedback, estimate modality contributions beyond confidence, and explicitly control the admission of historical information.

Despite progress in contextual modeling and multimodal fusion, existing approaches do not always coordinate history admission with modality contribution assessment. Three limitations are particularly relevant. First, methods that determine modality weights before historical feedback may fail to reassess contributions after context changes the current representations. Second, confidence-based routing can assign high weights to confident but redundant modalities, because confidence does not directly measure their marginal predictive contribution. Third, relevance-based history attention selects related utterances but, without an explicit abstention mechanism, may still introduce unhelpful or outdated emotional information. Consequently, modality dominance and harmful historical updates can persist even when fusion is adaptive. These limitations motivate feedback-conditioned reassessment, contribution-supervised bounded routing, and explicit history abstention.

To address these challenges, we develop MHnoU with three coordinated mechanisms. Feedback-Conditioned Context Reassessment (FCCR) uses current modality representations to retrieve historical information, incorporates admitted feedback, and reassesses modality contributions from the updated representations. Shapley-Anchored Bounded Evidence Routing (SABER) trains a contribution predictor using sample-level Shapley targets derived from predictive utilities across modality subsets. The predicted contribution scores are centered and normalized, then mapped to bounded deviations from uniform modality weights, constraining excessive reliance on a single modality. Null-Candidate History Abstention (NCHA) introduces a learned null candidate that competes with real historical candidates during attention. A history utility estimate calibrates this competition and determines a separate keep gate that controls whether historical feedback is applied. Rejected updates are blocked from the associated history paths, while admitted feedback informs subsequent modality routing. Together, FCCR, SABER, and NCHA are designed to make contextual updates selective and modality fusion sensitive to their resulting contributions.

These mechanisms coordinate history admission and modality routing within MHnoU, but an additional perspective can further support the selection of useful multimodal evidence. When textual content is ambiguous or conflicts with vocal and facial cues, a joint multimodal judgment may underuse informative nonverbal signals. An independently trained audio–visual predictor provides a

complementary assessment without relying on text. Its output, however, should be admitted selectively because nonverbal predictions can also be uncertain or reflect class-frequency preferences. We therefore introduce cRBEF as a parallel complementary evidence model, designed to assess when nonverbal support should influence a multimodal prediction and how strongly that support should be applied.

Specifically, cRBEF combines a text–audio–visual predictor with a separately trained joint audio–visual predictor. The audio–visual branch integrates global and local acoustic representations with visual features to produce a complementary emotion distribution. It constructs class-specific evidence by comparing audio–visual log probabilities with log class priors, expressing support relative to the prior distribution. A reliability gate uses prediction confidence, uncertainty, evidence strength, and agreement between the two predictors to regulate this evidence. The gated evidence then adjusts the text–audio–visual log probabilities before normalization. This mechanism aims to retain useful nonverbal corrections while reducing unreliable updates, providing a complementary route for evidence selection alongside the contribution-guided routing and selective contextual processing of MHnoU.

Building on these complementary roles, we propose ReCoMER (Reliability and Context Modeling for Emotion Recognition), a unified framework comprising MHnoU and cRBEF as two parallel models. MHnoU provides selective contextual reasoning through FCCR, SABER, and NCHA, while cRBEF supplies a reliability-regulated prediction informed by complementary audio–visual evidence. ReCoMER combines their probability distributions through bounded local class-wise correction, starting from their average prediction and using auxiliary reference predictions to inform limited adjustments. The framework is designed to coordinate evidence reliability, constrained modality weighting, and selective historical updates, with the aim of improving recognition while limiting harmful corrections.

The main contributions are as follows:

- We propose ReCoMER, a unified framework that combines selective contextual reasoning and complementary evidence assessment through two parallel models and bounded prediction fusion.
- We develop MHnoU, integrating FCCR, SABER, and NCHA to reassess modality contributions after historical feedback, constrain contribution-guided modality weights, and selectively accept historical updates.
- We incorporate cRBEF, which combines text–audio–visual and joint audio–visual predictors through prior-relative evidence and reliability gating, aiming to improve the use of informative nonverbal cues while limiting unreliable corrections.

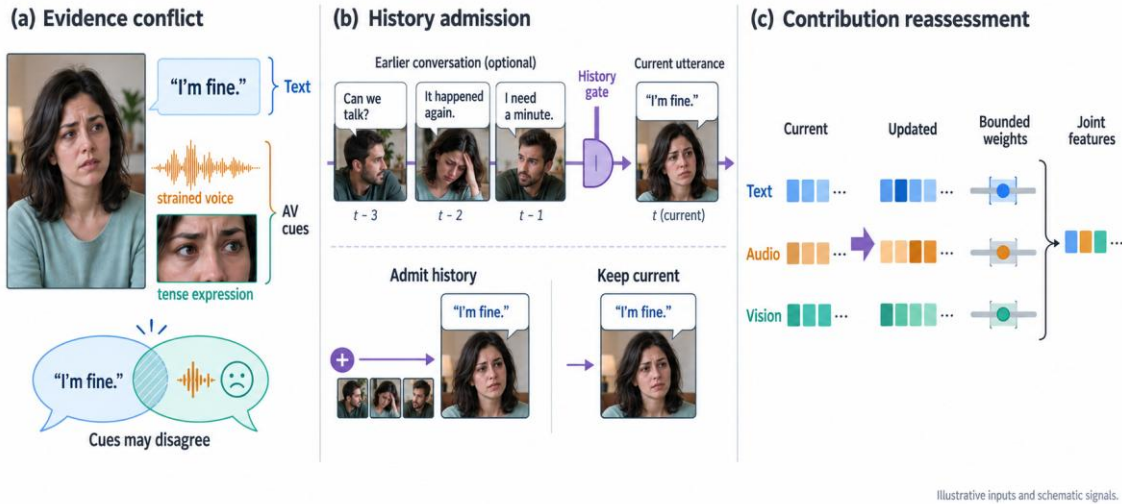


Figure 1. Motivation for selective context and modality evidence control.

2 Related Work

2.1. Multimodal Emotion Recognition in Conversations

Research on multimodal emotion recognition in conversations has sought to integrate dialogue dependencies with interactions among textual, acoustic, and visual cues. Recurrent approaches, exemplified by DialogueRNN [1], track conversational and speaker states to represent emotional dynamics. Graph-based approaches make utterance and modality relations explicit: MMGCN [2] connects contextual and cross-modal information within a multimodal graph, while M³Net [3] extends relational modeling to higher-order dependencies and multiple frequency components. These approaches differ in how they organize and propagate information, with recurrent models tracking conversational states and graph models explicitly representing relations among utterances and modalities.

Context selection further determines which information participates in these representations. Attention and gating provide one approach, as in ECERC [4], which models evidence–cause interactions and gates the resulting features. Relevance filtering and sparse graph routing provide another, as in HiRoC [5], which restricts historical interactions before aggregating them. These methods establish selective context integration as part of conversational modeling. DF-ERC [6] further regulates contextual integration using consistency among modality representations. Within this line of work, MHnoU uses selective history feedback and learns an empty-history candidate that can abstain from the additional update on a per-sample basis.

2.2. Modality Imbalance and Contribution Estimation

Multimodal learning can be imbalanced when modalities learn at different rates or one modality dominates joint optimization. Optimization-based approaches address this problem by regulating how individual modalities are trained. Gradient modulation [7] adjusts modality-specific updates using prediction scores that reflect the relative predictive strength of the modalities, while prototype-based rebalancing [8] provides direct learning signals for slower-learning modalities. PaSE [9] combines prototype alignment with Shapley-based gradient modulation. These methods act on learning dynamics, with modality assessments guiding optimization. Their role differs from determining how much each learned representation should contribute to the prediction of a particular sample.

Adaptive fusion addresses the latter problem by learning sample-dependent modality weights. A key distinction is the target used to supervise those weights. DF-ERC [6] uses unimodal true-class probabilities, while EMOE [10] derives importance targets from unimodal prediction errors and regularizes its router to limit modality dominance. Both assess how well a modality predicts on its own. Contribution estimates based on modality subsets instead assess changes in predictive utility when modalities are combined. DREAM [11] uses Shapley-based prediction-shift and prediction-accuracy contribution estimates, and distills the latter into fusion weights, connecting contribution assessment directly to adaptive fusion. Building on contribution-based weighting, MHnoU estimates modality contributions after history feedback and uses them to make bounded weight adjustments. This ordering differs from DF-ERC's modality weighting before context refusion [6].

2.3. Reliability-Aware Multimodal Fusion

Reliability-aware fusion regulates the influence of uncertain or conflicting information at different stages. At the representation level, EAU [12] models unimodal aleatoric uncertainty with probability distributions and uses estimated variances to weight features during fusion. This approach characterizes uncertainty within learned representations. At the prediction level, reliability controls the combination of already formed outputs. In bioacoustic classification, FINCH [13] combines acoustic predictions with spatiotemporal priors in log-probability space, using prediction statistics and contextual metadata to determine the supplementary weight. For conversational emotion recognition, HiRoC [5] uses disagreement between textual and nonverbal predictions to gate separate acoustic and visual residual logits. These methods illustrate that reliability can depend on both individual prediction statistics and relationships among predictions.

Our cRBEF belongs to this prediction-level line of work. It uses the posterior of a separately trained audio-visual predictor as supplementary evidence for its TAV prediction. The supplementary term is the log ratio of this posterior to the class prior, and its influence is controlled using confidence statistics, evidence strength, and predicted-class agreement. This differs from learning corrective residual logits within the conversational model: the correction is derived from a separately trained nonverbal posterior with an explicit prior reference. MHnoU and cRBEF each produce a complete class distribution. ReCoMER uses their average as the baseline and applies a bounded local_current correction using the TAV and no-history reference distributions together with modality evidence.

Recent conversation models also provide complementary context baselines. DialogueCRN [14] performs iterative contextual reasoning over retrieved emotional clues, DialogueGCN [15] models self- and inter-speaker dependencies as a graph, and COSMIC [16] injects commonsense relations such as intents, events and mental states. TelME [21] uses language-led cross-modal distillation and shifting fusion to strengthen non-verbal students. These methods improve contextual representation or cross-modal transfer, whereas ReCoMER focuses on the joint control of history admission, contribution supervision and an independent prediction-level expert.

3 Method

3.1 Task formulation and framework overview

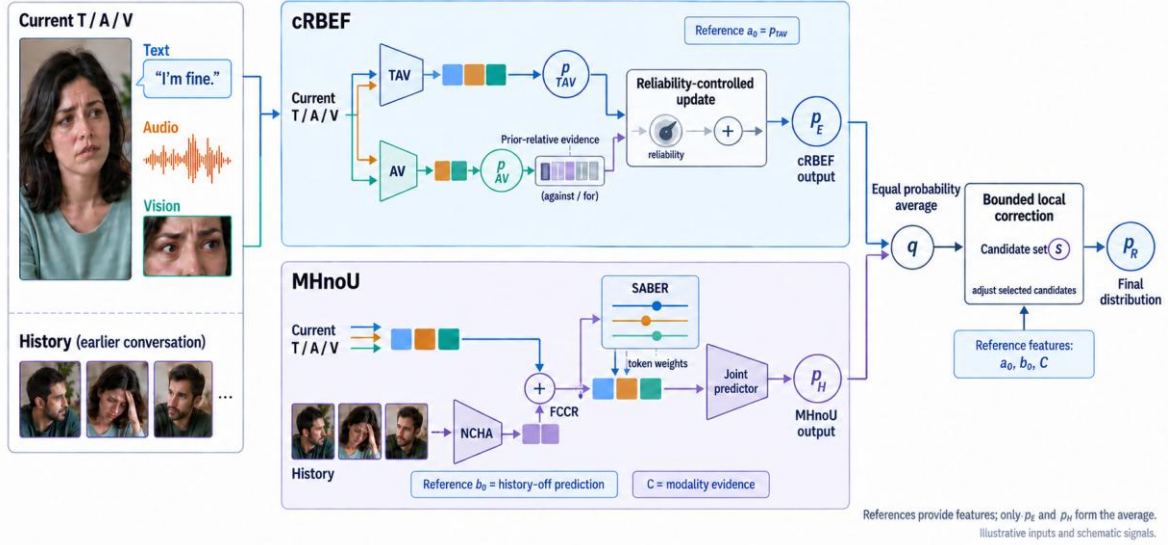


Figure 2. ReCoMER framework overview.

Given an utterance at turn t , we predict its emotion label y from current textual, acoustic, and visual features and a bounded history window. The modality set is $\{T, A, V\}$, and K denotes the number of emotion classes. The current mask M_t and history-slot masks identify available inputs. MHnoU applies modality-specific encoders, with each encoder shared between current and historical features:

$$z_t^m = E_m(x_t^m), \quad h_j^m = E_m(x_j^m), \quad m \in \mathcal{M} = \{T, A, V\}.$$

Equation 1

ReCoMER contains two parallel prediction models and an outer fusion module. MHnoU coordinates Feedback-Conditioned Context Reassessment (FCCR), Shapley-Anchored Bounded Evidence Routing (SABER), and Null-Candidate History Abstention (NCHA). Its forward path first assesses history admission, incorporates admitted feedback, and routes the resulting modality representations. Independently, cRBEF combines a text–audio–visual predictor with a joint audio–visual predictor. Bounded class-wise correction then combines their complete predictions. The two models retain their own feature representations and are aligned by utterance identifiers and class order.

3.2 Feedback Conditioned Context Reassessment

MHnoU: update first, then reassess

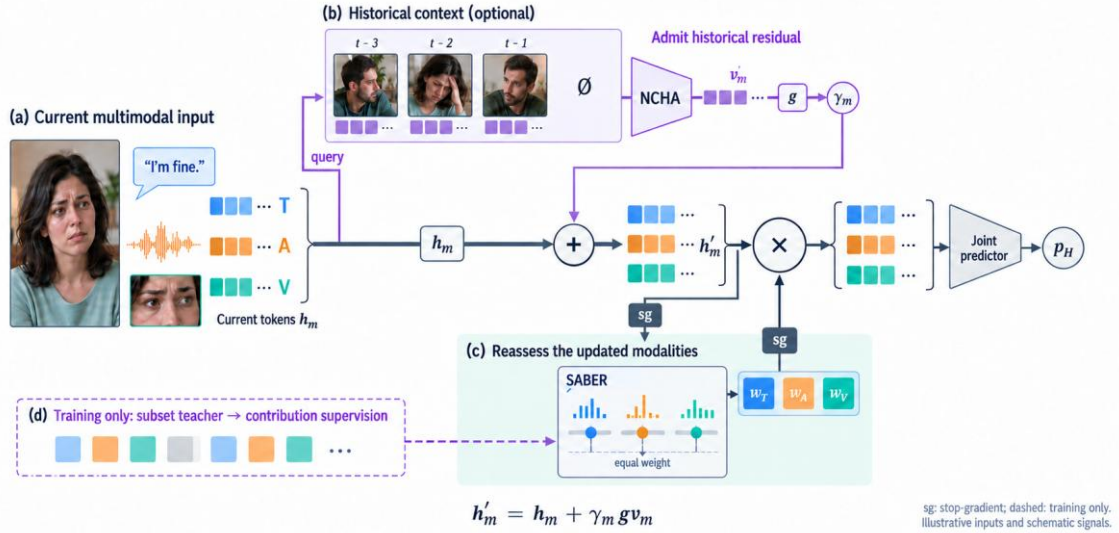


Figure 3. MHnoU history feedback, contribution routing and history admission.

A standard history-attention block has a concrete failure mode: it assigns mass to every valid-looking slot even when a previous utterance is irrelevant or only partially observed. FCCR is introduced to make history a conditional correction rather than an unconditional input. FCCR allows historical evidence to affect modality assessment before the final joint prediction. The implementation first encodes the current utterance and up to three preceding slots with the same modality-specific encoder, so a historical text vector is mapped into the same 192-dimensional space as the current text vector, and the same parameter sharing is used for audio and visual features. For each modality m , the current embedding queries only the valid historical slots through scaled dot-product attention. The history mask and the per-modality availability mask are multiplied before attention; therefore a padded slot, a missing modality, or a history item that does not contain modality m cannot contribute a value vector. We write this feedback-conditioned update in Eq. (2): the first term is the current modality state, the second term is the masked history summary, and γ_m controls the residual strength.

$$s_{ij}^m = \frac{\langle W_q^m z_t^m, h_j^m \rangle}{\sqrt{d}}, \quad \bar{h}_t^m = \sum_{j \in \mathcal{H}_t} \alpha_{ij}^m h_j^m, \\ \tilde{z}_t^m = z_t^m + g_t \gamma_m \bar{h}_t^m.$$

Equation 2

The design still needs a constraint because an unconstrained feedback residual could amplify a noisy history representation. The scalar feedback scale γ_m is learned separately for text, audio, and vision and is initialized to zero. The historical summary contains only real, valid historical values; the null candidate used by NCHA contributes a score but no value vector. After the history decision, each updated modality representation is passed to an auxiliary solo head. They provide modality-wise evidence that the router can inspect after feedback has changed the representation. The updated representations therefore follow a closed loop: history is selected and admitted first, the current modality states are updated, and the updated states are reassessed before the joint decision. Eq. (3) therefore describes a single reassessment step: auxiliary modality logits are computed after feedback, and the resulting evidence is passed to the router before joint prediction.

$$p^H = \text{softmax}(F_{\text{joint}}(\{\text{sg}(w_m) \tilde{z}_t^m\}_{m \in \mathcal{M}}, M_t)).$$

Equation 3

A second failure mode is that missing tokens can distort a joint head if masking is applied only at the input boundary. Here sg denotes stop-gradient. The joint predictor receives the three current modality tokens after feedback and routing. It uses a two-layer six-head Transformer encoder with feed-forward width 576, masked attention, masked mean pooling, and a seven-class output layer

for M3ED. The modality mask is carried into both attention and pooling; unavailable tokens do not dilute the representation or receive an artificial uniform contribution.

3.3 Shapley Anchored Bounded Evidence Routing

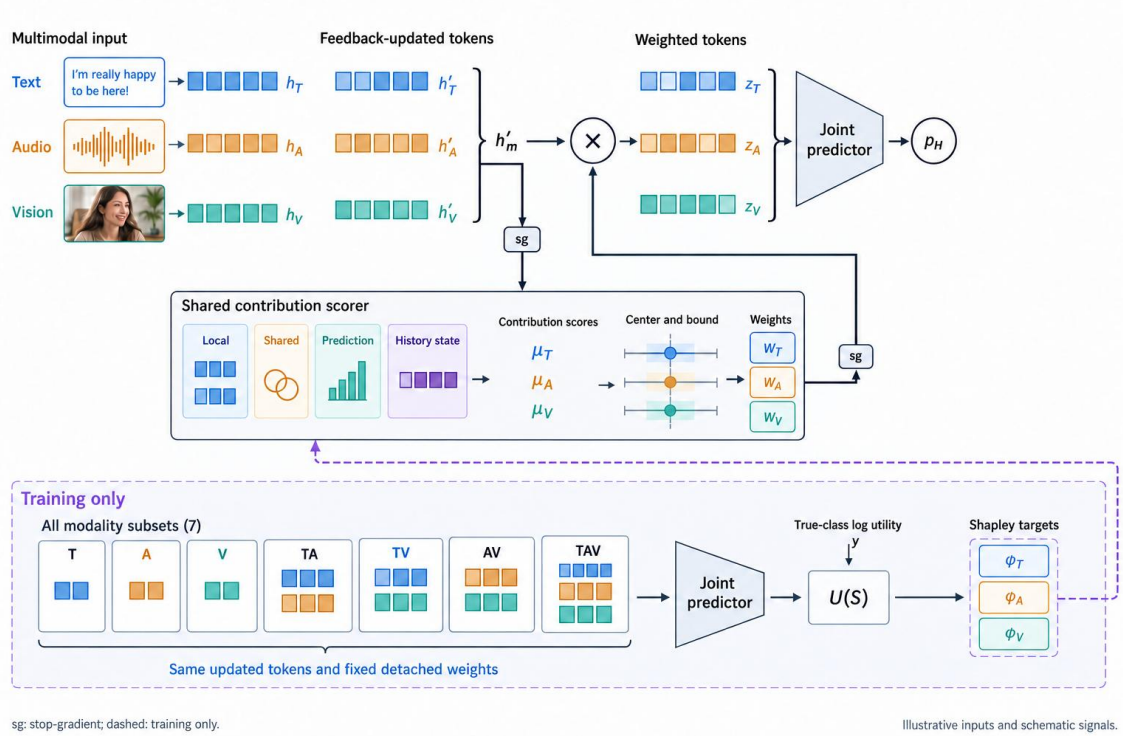


Figure 4. SABER contribution supervision and bounded routing.

Uniform averaging or a free learned gate has a specific weakness: it cannot explain why a modality is useful for one utterance but unreliable for another. SABER is designed to replace this opaque preference with evidence anchored to sample-level utility. SABER estimates modality contribution from both local evidence and shared cross-modal context. Each updated modality embedding is detached and projected by a modality-specific LayerNorm-linear-GELU block to 48 dimensions. A masked mean produces the shared context. For modality m , the scorer receives the local projection, the shared context, their difference, and a detached evidence vector. In the M3ED classification setting, this evidence vector contains the class probability distribution, maximum probability, normalized entropy, top-two margin, centered-logit scale, disagreement with the available-modality mean prediction, top-class agreement, and a modality-presence flag. Four detached history-state values can additionally expose the history keep probability, the applied admission value, whether history exists, and the mean null-attention mass.

To prevent the router target from becoming a hand-written heuristic, we define a measurable subset utility. The training target is obtained from the current joint predictor rather than from a hand-written modality importance rule. For each training sample, the teacher evaluates all seven nonempty subsets of $\{T, A, V\}$. Each subset uses the same feedback-updated current tokens, the same closed-loop history decision, and detached deployment weights; raw history tokens are not appended to the teacher input. For classification, the utility of a subset is the log probability assigned to the ground-truth class, while the empty-set utility is defined as zero. This makes the target sample-specific: a modality is valuable when adding it improves the class probability for that utterance, not merely because it is globally informative on average. Equation (4) makes this definition explicit: the utility of a subset is the log probability of the observed class, with the empty utility fixed to zero.

$$U_t(S) = \log p_{t, y_t}(S), \quad U_t(\emptyset) = 0.$$

Equation 4

The raw subset utilities still have incomparable scales across samples, so the Shapley target must be normalized before it supervises the router. The three-player Shapley value averages marginal utility changes over subsets. For modality m , the target sums the marginal gain obtained when m is added to every subset that does not contain m , weighted by the corresponding factorial coefficient. The result is a three-element vector for each utterance. It is standardized within the sample using the mean and sample standard deviation of the three values before supervising μ . Standardization removes the arbitrary absolute scale of the log probabilities while preserving which modality is relatively more useful for that sample. Equation (5) then converts these marginal gains into a standardized three-component supervision target.

$$\phi_{tm} = \sum_{S \subseteq \mathcal{M} \setminus \{m\}} \frac{|S|! (2 - |S|)!}{3!} [U_t(S \cup \{m\}) - U_t(S)],$$

$$\hat{\phi}_{tm} = \frac{\phi_{tm} - \text{mean}_r(\phi_{tr})}{\max(\text{sd}_r(\phi_{tr}), 10^{-6})}.$$

Equation 5

The resulting scores also require a deployment constraint: unrestricted weights could let one noisy modality dominate or assign negative mass to a present modality. The detached Shapley estimates are converted into bounded deployment weights rather than being used as unrestricted multiplicative gains. The implementation applies a temperature-scaled softmax-like mapping with $\lambda=0.3$, followed by nonnegative clipping and renormalization over the available modalities. In the three-modality case, the nominal weights remain in a bounded interval of approximately 0.233 to 0.433. Missing-modality masks are applied independently of the three-way evidence calculation, so an absent modality receives no token mass even if a numerical scorer produces a finite value. Equation (6) maps the standardized estimates to bounded weights and keeps the support of the available-modality mask separate from the learned preference.

$$r_m = \mu_m - \frac{1}{3} \sum_{r \in \mathcal{M}} \mu_r,$$

$$w_m = \frac{1}{3} \left(1 + \lambda \frac{r_m}{\max(\max_r |r_r|, 10^{-6})} \right).$$

Equation 6

This bound is necessary for an interpretable ablation and for stable inference. Equal scores recover uniform weighting for ablation. The bounded route is therefore not a replacement for the current MHnoU predictor; it is a constrained evidence allocation layer placed between modality encoding and joint prediction. Inference uses the saved evidence-router parameters with no subset enumeration. The exact Shapley computation is used only to construct training targets, whereas deployment requires one forward pass through the local scorer.

3.4 Null Candidate History Abstention

NCHA: soft weighting, hard admission

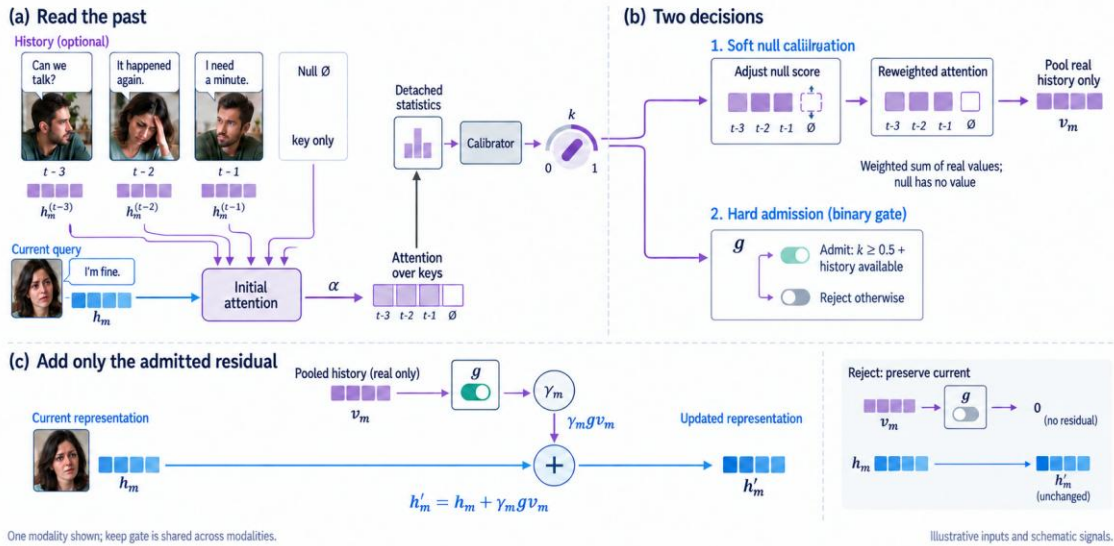


Figure 5. NCHA null-candidate history abstention.

Ordinary attention has no abstention option: even when all history is harmful, it still distributes probability over available slots. NCHA is introduced to make the model decide whether a historical update should exist at all. NCHA separates the selection of historical content from the decision to admit its update. The first attention pass estimates which real historical slots are relevant for

each modality. A second pass adds one learned null candidate per modality and lets that candidate compete with valid history. Before feedback, a detached 24-dimensional feature vector summarizes the evidence available to the admission decision. It includes current-utterance entropy, top-two margin, and modality presence for the three modalities; history coverage; current-versus-history prediction divergence and history entropy; speaker-related availability; and the existence and coverage of history. The three reserved text-similarity positions are zero in the base configuration.

$$k_t = \sigma(u_t/T_h), \quad b_t = \beta[\log(1 - k_t) - \log k_t],$$

$$g_t = \mathbb{I}[k_t \geq \theta] \mathbb{I}[\mathcal{H}_t \text{ has valid slots}].$$

Equation 7

The null candidate alone is not sufficient, because a soft null score does not define a stable deployment decision. The current implementation uses $T_h=1$, $\beta=1$, and a hard threshold $\theta=0.5$. Logarithm arguments are clamped at 10^{-5} before the keep probability is converted into a null-candidate bias. A high keep probability lowers the relative null score and allows real history to compete normally; a low keep probability raises the null score and makes abstention likely. The model can therefore select one historical slot more strongly than another while still rejecting the whole historical update when its expected utility is low. Equation (8) shows the second attention normalization: real history and the null candidate compete, but only real history contributes a value vector.

$$s_{t\emptyset}^m = \frac{\langle W_q^m z_t^m, n_m \rangle}{\sqrt{d}} + b_t,$$

$$\alpha_t^m = \text{softmax}([s_{tj}^m]_{j \in \mathcal{H}_t}, s_{t\emptyset}^m).$$

Equation 8

A soft null mass also needs a structural constraint; otherwise rejected history could re-enter through another memory path. Here n_m is the learned null embedding, and invalid historical scores are masked before normalization. The null mass reduces the total attention assigned to real history, but it does not create a fake feature vector that is later added to the current state. The null represents abstention, whereas a real slot represents a concrete previous utterance. The same admission value is also used by the optional memory path, preventing rejected history from bypassing NCHA through a second route. Hard rejection is determined by the keep threshold, not simply by whether the null candidate has the largest softmax probability. Equation (9) expresses the straight-through hard decision used during training and the binary decision used at inference.

$$g_t^{\text{ST}} = g_t + k_t - \text{sg}(k_t).$$

Equation 9

The hard inference rule is therefore a deliberate safety boundary, not an additional prediction target. At inference, the binary gate is used directly. If a sample is rejected, the historical residual in Eq. (2) is set to zero for the whole utterance, while the current modality encodings remain available. Because the closed-loop joint head receives no raw history tokens, rejected history cannot re-enter through an auxiliary token path. No history label is required at test time: the decision is based only on detached uncertainty, margin, availability, divergence, and speaker-state features computed from the current input and its available history.

3.5 Complementary evidence modeling with cRBEF

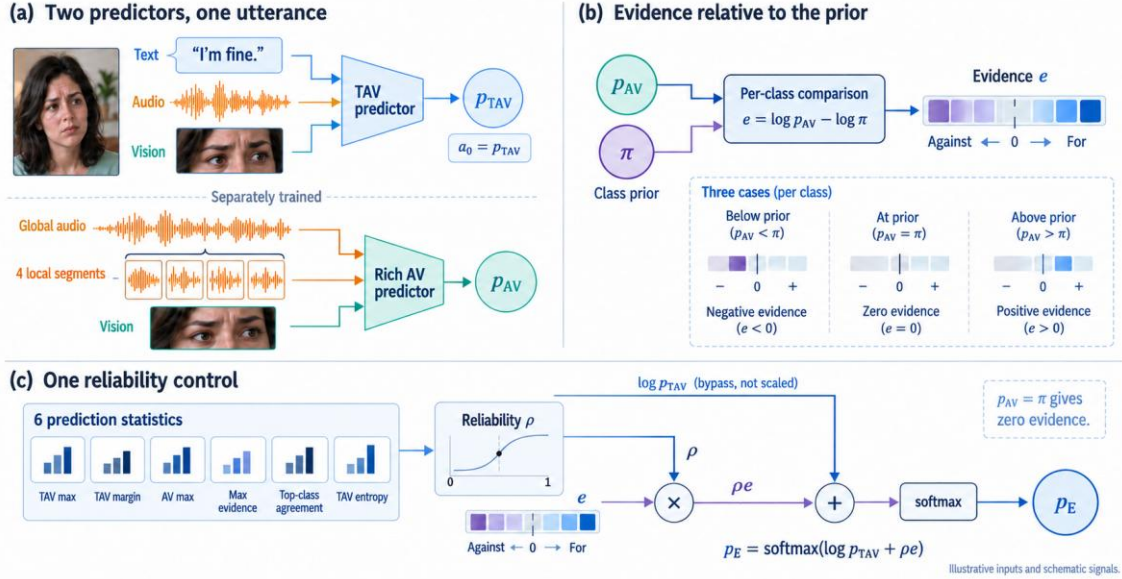


Figure 6. cRBEF prior-relative audio-visual evidence and reliability gating.

A single MHnoU predictor can still be confidently wrong when lexical context dominates a nonverbal cue. cRBEF is introduced as an independent external expert so that the final system receives a second, differently constructed opinion. Within ReCoMER, cRBEF complements feedback-conditioned history routing with an independent assessment of multimodal evidence. The expert is loaded as a frozen external module and does not share MHnoU encoders or router parameters. It contains a text–audio–visual (TAV) predictor, a separate joint audio–visual (AV) predictor, and a six-input reliability gate. TAV supplies the all-modality reference. The AV branch is deliberately independent of text and uses global and local acoustic summaries together with visual features.

The external branch must first be kept structurally independent; otherwise the proposed second opinion would merely duplicate MHnoU features. The TAV predictor consumes the original cRBEF feature contract: a 4096-dimensional text vector, a 1024-dimensional audio vector, and a 342-dimensional visual vector. Audio and vision are independently projected to 256 dimensions and layer-normalized. Binary presence flags mask the projected audio and visual features before they are concatenated with text, giving a 4608-dimensional input. A linear layer reduces this input to 128 dimensions, followed by GELU, dropout with probability 0.2, layer normalization, a seven-class linear head, and a learned scalar logit-temperature parameter implemented through tanh.

The AV branch remains text-free to preserve independent nonverbal evidence. The AV predictor uses its own 1024-dimensional global acoustic summary and four 1024-dimensional local acoustic summaries. The local summaries are flattened to 4096 dimensions. Global audio, local audio, and vision are each projected to 256 dimensions and layer-normalized, then concatenated into a 768-dimensional input to a 128-dimensional classifier with the same GELU, dropout, normalization, and logit-scaling structure as the TAV branch.

A naive probability average cannot express class-specific support, because the AV branch may support one class while contradicting another. Let p^{TAV} and p^{AV} denote the two probability distributions and π the stored training class prior. For each class c , cRBEF forms prior-relative evidence $e_c = \log p^{\text{AV}}_c - \log \pi_c$. Positive evidence means that the AV branch supports class c more strongly than its prior frequency would suggest; negative evidence means that the AV branch is less supportive than the prior. The gate scales the complete class-wise evidence vector, so several classes can be increased or decreased together. Equation (10) uses this evidence to form a class-wise log-probability correction rather than a global scalar average.

$$e_c = \log p_c^{\text{AV}} - \log \pi_c, \quad \rho = \sigma(v^T s + b),$$

$$p_c^E = \frac{\exp(\log p_c^{\text{TAV}} + \rho e_c)}{\sum_{r=1}^K \exp(\log p_r^{\text{TAV}} + \rho e_r)}.$$

Equation 10

The gate is therefore constrained to use reliability statistics rather than raw feature concatenation. The gate input s has six components implemented exactly in the expert adapter: the maximum TAV probability, its top-two margin, the maximum AV

probability, the strongest prior-relative AV evidence, a binary indicator of top-class agreement, and the unnormalized TAV entropy. A learned linear map from these six statistics to one logit, followed by a sigmoid, produces an utterance-specific gate ρ in (0,1).

The final evidence equation makes the intended limiting cases explicit. The gated evidence is added to the TAV log probabilities and normalized over classes to produce $p^E = \text{softmax}(\log p^{\text{TAV}} + \rho e)$. When ρ approaches zero, cRBEF reduces to the TAV reference; larger ρ applies a stronger prior-relative AV correction. Because e can be positive or negative, cRBEF can either strengthen or suppress a class.

3.6 Bounded local class wise fusion

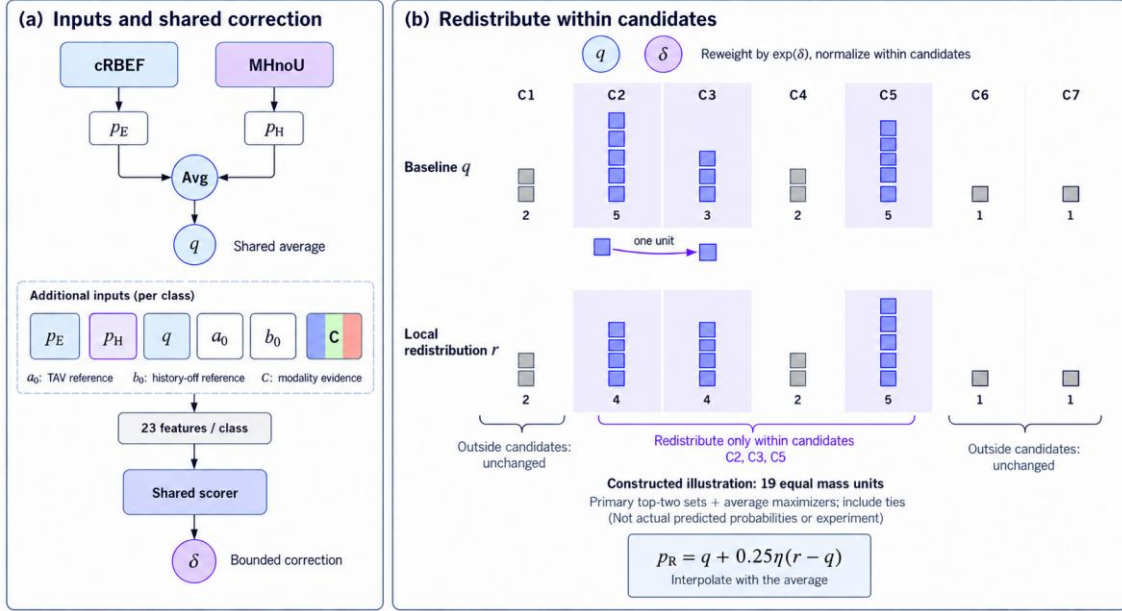


Figure 7. Bounded local class-wise fusion and equal-weight fallback.

After obtaining two complete peer predictions, simple averaging still has a concrete limitation: it cannot use the internal reference changes or the relative modality evidence to resolve a disagreement. The outer fusion uses complete predictions and two reference distributions. The four columns of P are ordered as $[p(\text{cRBEF}), p(\text{MHnoU}), p(\text{TAV}), p(\text{no-history MHnoU})]$. The first two define the peer baseline; the last two anchor changes from each internal reference. MHnoU and its no-history reference are generated from the same checkpoint, with the latter obtained by zeroing the history masks. Equation (11) records this equal-probability baseline explicitly so that the learned module has a transparent reference.

$$P = [a, b, a_0, b_0] = [p^E, p^H, p^{\text{TAV}}, p^{\text{H0}}], \quad q = (a + b)/2.$$

Equation 11

Because equal averaging is intentionally conservative, the corrector needs a separate evidence channel that explains a disagreement without changing either expert. The no-history MHnoU auxiliary logits provide additional relative evidence about the current modalities. The implementation first removes the class-wise mean logit within each modality and then subtracts the mean across available modalities. This produces C with one centered evidence vector for text, audio, and vision. C is not a second prediction and is not fed back into cRBEF; it records which modality is relatively more supportive of a class under the MHnoU checkpoint. P and C remain separate by design.

$$L_{mc}^0 = \ell_{mc}^0 - \frac{1}{K} \sum_{r=1}^K \ell_{mr}^0,$$

$$C_{mc} = M_m \left[L_{mc}^0 - \frac{\sum_{r \in \mathcal{M}} M_r L_{rc}^0}{\max(\sum_{r \in \mathcal{M}} M_r, 1)} \right].$$

Equation 12

An unconstrained MLP over concatenated predictions could learn arbitrary class transfers and overfit the fitting split. For each class, the local corrector receives 23 features. Six are the two primary probabilities, their logarithms, the equal-weight baseline probability, and its logarithm. Four global statistics describe the entropy and top-two margin of each primary prediction. Two indicators mark whether each primary assigns the class the highest probability. Eight reference features contain the two reference probabilities, their logarithms, the change from each reference to its corresponding complete prediction, and the L1-normalized versions of those changes. The final three features are the centered relative evidence for T, A, and V. Feature statistics are fitted only on the fusion set and clamped to 10^{-5} . Equation (13) standardizes these 23 features and produces a bounded class-wise residual score.

$$\delta_c = 1.5 \tanh(f_\psi(\text{clip}((x_c - \bar{x})/s_x, -10, 10))).$$

Equation 13

Even a well-trained scorer can be unsafe if it is allowed to move probability to every class. The candidate set restricts correction to classes that already have local support. It contains every class whose probability is at least the second-largest probability of either primary expert, together with the class that maximizes the equal-weight baseline; ties at the cutoff are retained. Inside the candidate set, the network's bounded residual is exponentiated and used to redistribute the candidate mass. Noncandidate probabilities are preserved at this stage. Equation (14) reallocates only the probability mass inside the candidate set.

$$B = \sum_{j \in \mathcal{K}} q_j, \quad r_c = \frac{B q_c \exp(\delta_c)}{\sum_{j \in \mathcal{K}} q_j \exp(\delta_j)}, \quad c \in \mathcal{K},$$

$$r_c = q_c, \quad c \notin \mathcal{K}.$$

Equation 14

A correction that can replace the baseline completely would also erase the reliable cases of the two experts. The final prediction interpolates between the corrected distribution and the equal-weight baseline. The current implementation uses a correction fraction of 0.25 and stores a selected η in the bundle; $\eta=0$ exactly recovers the baseline, while larger η exposes more of the learned correction. During fitting, the corrector is trained with negative log-likelihood, a preservation penalty on samples that the baseline already classifies correctly, and a small squared-residual regularizer. The preservation term limits unnecessary changes without assuming every baseline decision is correct. FIT and CAL are source- and sample-disjoint. Equation (15) interpolates the corrected distribution with the baseline using the selected η .

$$p^R = q + 0.25\eta(r - q), \quad \eta \in [0, 1].$$

Equation 15

The final constraint is a recoverable baseline: the fitted mechanism must be able to return exactly to equal averaging when evidence is insufficient. The last correction layer is zero-initialized, so an unfitted LocalCurrentFusion instance is exactly the equal-probability baseline. Candidate mass and noncandidate probabilities are preserved before the final 10^{-7} clamp and normalization. This initialization provides a safe fallback when the fitted corrector does not improve the calibration set.

3.7 Training and inference of ReCoMER

ReCoMER is trained and deployed as a complete framework consisting of MHnoU, cRBEF, and the bounded class-wise fusion module. Its current implementation uses staged optimization: the MHnoU component is trained for contextual reasoning and contribution estimation, the separately trained cRBEF expert is reused, and the outer corrector is fitted with both prediction models frozen. The component objective and the final fusion objective describe different stages of this overall procedure.

In the first stage, the MHnoU component combines joint classification supervision with standardized Shapley regression:

$$\mathcal{L}_{MH} = \text{CE}_{\omega, 0.05}(p^H, y) + 0.3 \text{MSE}(\mu, \hat{\phi}).$$

Equation 16

Cross-entropy uses label smoothing of 0.05 and class weights $\min(2, \sqrt{(N/(K - n_c))})$, with class counts n_c clamped to at least one. This stage trains the contextual prediction path and contribution router. NCHA receives task gradients through historical feedback and its straight-through gate. The active configuration adds no separate unimodal classification losses or cache-supervised history BCE term. Because the joint and deployed component outputs are identical, their consistency term is zero.

In the second stage, ReCoMER’s fused distribution p^R is the prediction being supervised. The four aligned distributions and modality-relative evidence in Section 3.6 supply the outer corrector’s inputs. MHnoU and cRBEF remain frozen, and only the corrector parameters are updated. The final fusion objective combines negative log-likelihood of the complete ReCoMER output, a preservation penalty on initially correct baseline predictions, and correction regularization:

$$\mathcal{L}_F = \frac{1}{N} \sum_{i=1}^N \left[-\log p_{i, y_i}^R + 2I_i \max(\log q_{i, y_i} - \log p_{i, y_i}^R, 0) \right] \\ + 0.01 \text{mean}_{i, c}(\delta_{ic}^2), \quad I_i = \mathbb{I}[\arg \max_c q_{ic} = y_i].$$

Equation 17

Here $\mathcal{L}_F$ denotes the objective for ReCoMER’s final fusion stage. Its preservation term discourages reductions in true-class confidence on correct baseline predictions without guaranteeing error-free updates. Fusion fitting and strength selection use sample- and source-disjoint FIT and CAL sets. The corrector is fitted with $\eta = 1$; η is then selected from $\{0, 0.25, 0.5, 1\}$ by the complete framework’s weighted F1 on CAL, with macro F1 and accuracy as tie-breakers. Source-group exclusion applies to task-head fitting and selection and does not eliminate prior supervision exposure in upstream feature models.

The current complete ReCoMER bundle implements seven-class M3ED recognition. Within it, MHnoU uses 192-dimensional embeddings, a two-layer six-head Transformer with feed-forward width 576, dropout 0.15, and up to three historical slots. Its text, audio, and vision input dimensions are 2560, 2048, and 342. cRBEF uses its original 4096-dimensional text, 1024-dimensional audio, 342-dimensional vision, and global/local acoustic summaries. MHnoU uses AdamW with learning rate 2×10^{-4} and weight decay 0.01; the outer corrector uses AdamW with learning rate 10^{-3} and weight decay 0.01 for 40 epochs. The saved framework includes both prediction models, the fusion parameters, and the selected correction strength.

At inference, ReCoMER receives MHnoU’s current/history inputs and the corresponding utterance’s original cRBEF features. MHnoU produces p^H and, from the same checkpoint with history masks disabled, $p^{\{H0\}}$ and auxiliary modality evidence. Independently, cRBEF produces p^E and its TAV reference. The framework aligns these outputs, constructs P and C , and applies the bounded correction and interpolation in Eqs. (11)–(15). The final emotion label is the highest-probability class of p^R , the fused ReCoMER distribution. No class or history labels enter this inference path. Application to a different class vocabulary requires compatible cRBEF predictors and a refitted outer fusion module.

4 Experiments

4.1 Experimental setup

We evaluated ReCoMER on multimodal conversational emotion recognition using four complementary evidence sources: a frozen formal test of the complete system, module-isolation studies for the three proposed components, corrected abstention experiments, and an exact-feature replication of the cRBEF expert. This separation was maintained throughout the analysis because the experiments use different data splits, numbers of random seeds and fusion scopes.

ReCoMER contains four functional parts. MHnoU is the history-aware main predictor. EvidenceRouter estimates sample-specific modality contributions from local modality evidence, cross-modal context and prediction statistics; it is evaluated in a frozen-backbone setting when used for module isolation. The history-abstention module lets real historical candidates compete with an explicit empty candidate and, when history is rejected, suppresses both the historical residual path and the raw historical-token path. cRBEF is an external class-probability expert. Its output is combined by the outer ReCoMER fusion module and is not fed back into the MHnoU modality router.

We used MELD, M3ED and IEMOCAP for classification and CMU-MOSEI for regression. Weighted F1 (WF1) was used for classification, with larger values indicating better performance; mean absolute error (MAE) was used for MOSEI, with smaller values indicating better performance. Innovation 1 and Innovation 2 used ten random seeds (7, 13, 17, 23, 29, 37, 43, 53, 71 and 101). Innovation 3 used three seeds under its corrected protocol. The frozen formal ReCoMER test consisted of three paired runs, with the cRBEF checkpoint fixed at seed 17. The exact-feature M3ED queue used ten seeds for the MHnoU controls and five seeds for the standalone cRBEF replication.

For the formal M3ED runs, the model used a 192-dimensional internal representation, dropout 0.15, AdamW with learning rate 2×10^{-4} and weight decay 0.01, batch size 256, warm-up followed by cosine scheduling, a maximum of 20 epochs and early stopping based on the selection split. The exact M3ED HuBERT pack contained 17,427 training samples, 2,821 validation samples and 4,201 test samples, with feature dimensions 768, 2,048 and 342 for text, audio and video, respectively. The remaining innovation-specific configurations followed their recorded run manifests. Test data were not used for module selection; the formal test was opened only after the available model and fusion controls had been frozen.

The ten-seed confidence intervals in the module-isolation studies are paired seed-level intervals on fixed validation protocols. The confidence intervals in the formal ReCoMER table are dialogue-bootstrap intervals over test conversations. These uncertainty definitions are not interchangeable.

4.2 Main comparison on the frozen M3ED test

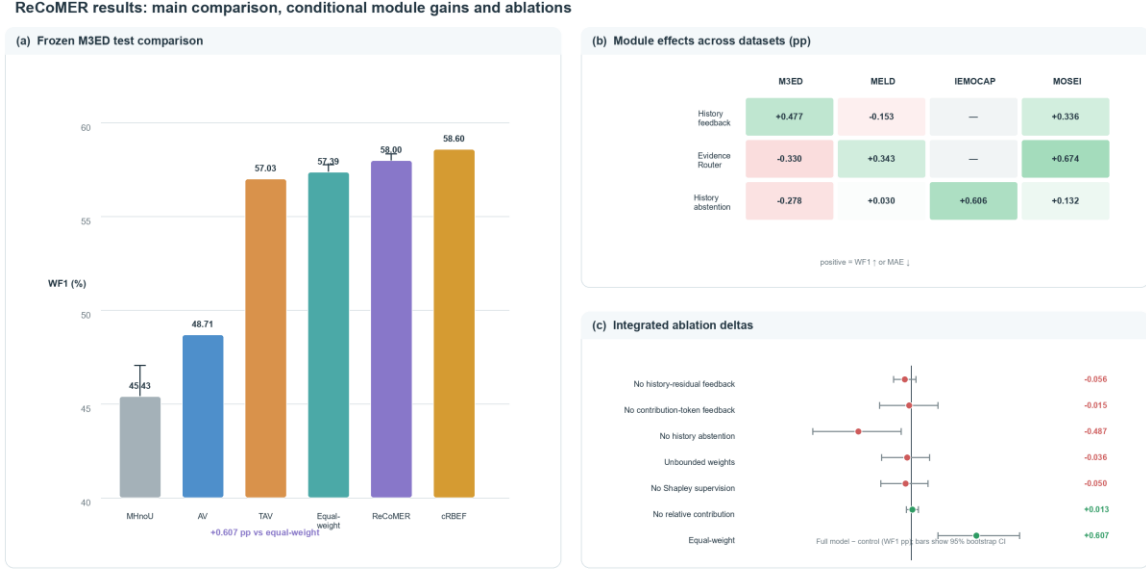


Figure 8. Formal M3ED comparison, conditional module effects and paired integrated ablations.

We first compared the complete ReCoMER system with its available fusion controls under the frozen M3ED test protocol. The results are shown in Table 1.

Table 1. Frozen M3ED formal test comparison. ** WF1 is reported in percent; values are higher-is-better. The cRBEF checkpoint was fixed at seed 17, whereas the ReCoMER and equal-weight values summarize the three paired runs.

Model or fusion rule	Test WF1 (%)
AV	48.71
TAV	57.03
Equal-weight fusion	57.39 ± 0.37
ReCoMER	**58.00 ± 0.34**
cRBEF (fixed seed 17)	58.60 ± 0.00

ReCoMER improved over the matched equal-weight control by 0.607 percentage points (dialogue-bootstrap 95% CI, 0.250–1.005). The learned outer fusion therefore provided a measurable gain over uniform combination. However, the fixed cRBEF expert remained slightly higher than the complete ReCoMER result. Thus, the present test supports the usefulness of external-expert fusion, but does not support the stronger claim that the learned fusion is better than every individual expert.

For context, the original M3ED paper reported a full-modality DialogueRNN WF1 of 51.66%. Because the current ReCoMER result uses different feature extraction and a different implementation protocol, this comparison is contextual rather than a strict same-protocol SOTA claim.

4.3 Innovation 1: historical feedback as conditional context compensation

We next asked whether historical feedback improves the current utterance prediction. To isolate this question, we compared history-enabled and no-history models under uniform modality weights. Table 2 reports the paired ten-seed effects.

Table 2. Historical feedback under the ten-seed validation protocols. ** Positive values indicate an improvement in WF1 or a reduction in MAE, according to the metric direction.

Dataset and comparison	Metric	Mean effect	95% CI	Same-direction seeds
M3ED: uniform_h – uniform_nh	WF1 ↑	+0.004774	[+0.001105, +0.008444]	7/10
MELD: uniform_h – uniform_nh	WF1 ↑	−0.001534	[−0.008234, +0.005165]	5/10
MOSEI: mosei_nh – mosei_h	MAE ↓	+0.003361	[+0.000021, +0.006701]	8/10

The effect of history was therefore conditional on the data regime. The clearest positive result occurred on MOSEI, where historical feedback reduced MAE. The M3ED effect was positive but did not reach the pre-specified 8/10 same-direction criterion, whereas the aggregate MELD interval crossed zero. The available front/late/closed-loop controls also did not establish an independent gain from the ordering “feedback followed by reassessment”: FULL – ONE_GATE was +0.000042, FULL – UNIFORM_QUERY was +0.000114 and FULL – GLOBAL_LATE was −0.000765, with all corresponding intervals overlapping zero. We therefore interpret the current evidence as conditional historical-feedback fusion rather than proof that a particular closed-loop order is universally superior.

4.4 Innovation 2: Shapley-anchored contribution routing

We then isolated the contribution-routing component. The encoder, history module, solo heads and joint head were frozen, while only the EvidenceRouter utility head was optimized on a fixed uniform backbone. The router received exact per-sample Shapley supervision together with the task objective and a pairwise ranking term. These results measure the independent routing effect and should not be confused with a final full-SABER test.

Table 3. EvidenceRouter under the fixed-backbone ten-seed protocol.

Dataset	Uniform	EvidenceRouter	Change	95% CI	Positive seeds
MOSEI MAE ↓	0.566134	0.559393	−0.006741	[−0.008536, −0.004946]	10/10
MELD WF1 ↑	0.577705	0.581131	+0.003426	[+0.000938, +0.005914]	8/10
M3ED WF1 ↑	0.602071	0.598774	−0.003298	[−0.005441, −0.001154]	2/10

EvidenceRouter consistently improved MOSEI MAE and produced a smaller positive effect on MELD WF1. In contrast, the M3ED E2 representation did not benefit from the router. The MOSEI result was also metric-dependent: MAE decreased, whereas RMSE and Pearson correlation did not improve in the same experiment. These observations indicate that Shapley-anchored routing can be useful when the modalities provide complementary evidence, but its benefit is not independent of the feature regime.

4.5 Innovation 3: sentence-level history abstention

The third study tested whether the model could compete between real historical candidates and an explicit empty candidate. B1 denotes the original empty-candidate design. E5 applies utility calibration before softmax normalization, E6 additionally uses semantic history features, and E7 replaces softmax with sparsemax. All four variants retain the same candidate set and are evaluated under the corrected history-path implementation.

Table 4. Corrected history-abstention variants. Values are mean ± population standard deviation over three seeds.**

Dataset	Metric	B1	E5	E6	E7
MELD	inner-dev WF1 ↑	0.5967 ± 0.0069	0.5970 ± 0.0090	0.5968 ± 0.0088	0.5957 ± 0.0091
M3ED	valid WF1 ↑	0.561040 ± 0.0012	0.558256 ± 0.0008	0.558256 ± 0.0008	0.558650 ± 0.0023
IEMOCAP	valid WF1 ↑	0.724835 ± 0.0022	0.730898 ± 0.0065	0.730898 ± 0.0065	0.730378 ± 0.0058
MOSEI	valid MAE ↓	0.563965 ± 0.0026	0.562650 ± 0.0028	0.562651 ± 0.0028	0.563299 ± 0.0028

Relative to B1, E5 improved IEMOCAP WF1 by 0.006063 and reduced MOSEI MAE by 0.001315. It remained essentially unchanged on MELD (+0.0003 WF1) and decreased M3ED WF1 by 0.002784. E6 was numerically indistinguishable from E5, and E7 showed no stable advantage over softmax. The corrected results therefore support a dataset-dependent abstention effect rather than universal improvement.

4.6 Integrated ablations and information-flow checks

The frozen ReCoMER test also enabled paired ablations of the integrated model. Table 5 reports the difference between the complete model and each control.

Table 5. Integrated ReCoMER ablations. Values are Full – control in percentage points of test WF1.**

Control	Full – control	Dialogue-bootstrap 95% CI
Equal-weight fusion	+0.607	[0.250, 1.005]
No relative contribution C	+0.013	[−0.043, 0.069]
No Shapley supervision	−0.050	[−0.281, 0.158]

Unbounded softmax weights	−0.036	[−0.272, 0.172]
No history abstention	−0.487	[−0.911, −0.091]
No contribution-token feedback	−0.015	[−0.290, 0.248]
No history-residual feedback	−0.056	[−0.160, 0.047]

The complete model was better than equal weighting, and removing history abstention was harmful under this test protocol. The intervals for relative-contribution, Shapley-supervision and bounded-weight controls overlapped zero; these comparisons are therefore reported as mechanism indications rather than independent significance claims.

The cRBEF/MHnoU splice passed the available identity and information-flow checks. Replacing cRBEF changed the expert and final fused probabilities while leaving MHnoU probabilities and its modality-relative contribution unchanged. This supports the interpretation of cRBEF as an external probability expert rather than an accidental input to the MHnoU router. The CR17 adapter qualification also produced zero hidden replay error on 4,201 test rows.

4.7 Exact-feature replication and mechanism diagnostics



Figure 9. Exact-feature M3ED controls and cRBEF replication.

To separate model behavior from feature-protocol effects, we ran an additional exact M3ED queue using the official train/validation/test split and the exact HuBERT feature pack. The queue was deliberately labeled as a supplementary protocol because its five arms were MHnoU controls with the three integrated innovation flags disabled.

Table 6. Exact-feature M3ED ten-seed test controls.

Arm	Test WF1
uniform_nohist (B0)	0.535269 ± 0.005824
uniform_history	0.534562 ± 0.004858
evidence_closed	0.537279 ± 0.004118
evidence_mpath	0.539917 ± 0.006380
uniform_abstain	0.533945 ± 0.004854

Under the same exact-feature protocol, the best arm was evidence_mpath, whereas B0 reached 0.535269. These values are useful for controlled exact-feature comparisons but should not be merged with the 58.00% formal ReCoMER result, which uses the complete model and a different feature protocol.

We also retrained the cRBEF-style evidence heads on the exact T/A/V features. Gate parameters were selected on the validation split and frozen before test evaluation. The five-seed test WF1 values were 0.421594 ± 0.002423 for T-only, 0.531980 ± 0.001816 for AV-only, 0.528735 ± 0.005345 for ungated TAV, 0.526567 ± 0.005946 for log-evidence fusion and 0.548599 ± 0.004454 for gated CRBEF. This replication confirms that the expert-fusion arithmetic remains functional under the exact feature pack; it is not a replacement for the frozen formal ReCoMER result.

Finally, the ten-seed mechanism diagnostic on the evidence_mpath checkpoints measured history availability, keep probability, hard keep rate, abstention mass and history override sensitivity. On the exact test split, history was available for 0.952868 of samples, the mean keep probability was 0.950279, the hard keep rate was 0.952868, the mean abstention mass was 0.202451 and forcing the

history decision changed deployed logits by 0.046409 on average. These values demonstrate that the history path participates in deployment decisions, but they do not by themselves establish a performance gain.

4.8 Sensitivity, failure boundaries and reproducibility



Figure 10. Frozen cRBEF gate-formula sensitivity; this is a discrete formula robustness check rather than a continuous hyperparameter sweep.

The results revealed three scope conditions. First, M3ED is strong at the utterance level under the tested representation, so history feedback and contribution routing can have small or negative marginal effects. Second, MELD exposes a continuation-versus-transition trade-off: historical context can help continuation utterances but can also propagate an incorrect prior state across an emotional transition. Third, MOSEI benefits most consistently from historical compensation, but the direction is metric-specific and should not be generalized from MAE to every regression or classification measure.

The complete run package records model hashes, checkpoints, per-seed histories, Shapley diagnostics, modality weights, history decisions, token masks and split guards. The exact-feature queue provides a separate B0 and mechanism reference. The following analyses remain supplementary or pending rather than being silently inferred: a complete four-dataset B0 matrix, same-label versus different-label subgroup results, a full three-seed diagnostic matrix for every abstention variant, a train-only utility-cache comparison and a complete same-protocol external SOTA table.

4.9 Summary of experimental claims

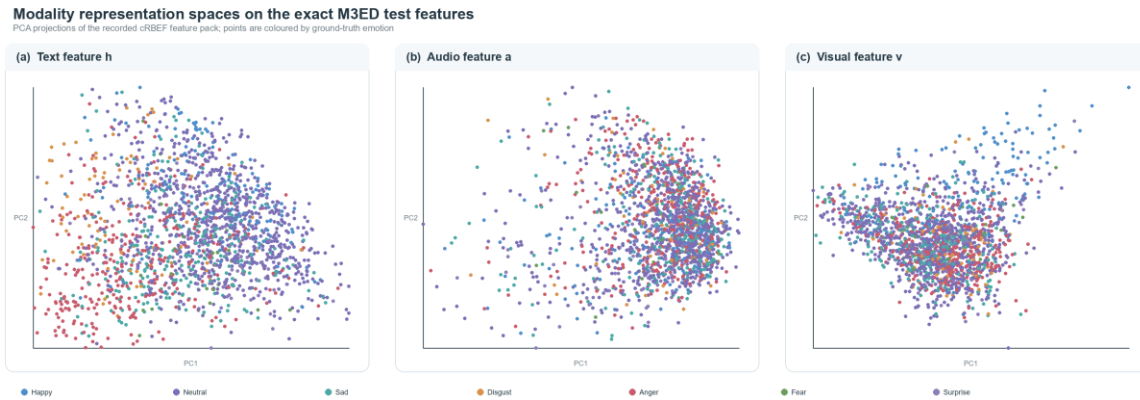


Figure 11. PCA projections of exact M3ED test features for text, audio and vision, colored by ground-truth emotion.



Figure 12. Mean row-normalized confusion structure for MHnoU, cRBEF and ReCoMER across the three formal M3ED test seeds.

The experiments support four bounded conclusions. First, ReCoMER improved over equal-weight fusion on the frozen M3ED formal test, although the fixed cRBEF expert remained slightly stronger. Second, historical feedback can compensate for weak utterance-level evidence, most clearly on MOSEI, but its effect is conditional on context. Third, Shapley-anchored contribution routing and sentence-level history abstention showed positive effects on selected datasets but did not transfer uniformly to M3ED. Fourth, the information-flow audit supports cRBEF as an external expert in the outer fusion. Together, these results establish a reproducible and interpretable fusion framework while making its data-dependent failure boundaries explicit.

5 Conclusion

Multimodal conversational emotion recognition is difficult because the reliability of historical context and individual modalities changes from one utterance to another. ReCoMER addresses this problem with a structured conditional-fusion framework: MHnoU provides the history-aware main prediction, EvidenceRouter estimates sample-specific modality contributions with Shapley-anchored supervision, the history module can abstain through an explicit empty candidate, and cRBEF supplies an external probability expert that is combined only at the outer fusion stage. This separation makes the information flow auditable and allows the contribution of each component to be tested independently.

Under the frozen formal M3ED test protocol, ReCoMER achieved 58.00% WF1 and improved over matched equal-weight fusion by 0.607 percentage points. The module-isolation studies further showed conditional benefits: historical feedback produced its clearest gain on MOSEI, EvidenceRouter improved MOSEI and MELD under the fixed-backbone protocol, and the corrected abstention variants improved IEMOCAP and MOSEI relative to the original empty-candidate design. The cRBEF/MHnoU information-flow checks also confirmed that cRBEF acts as an external expert rather than an unintended input to the MHnoU modality router.

These results define the scope of the contribution. The fixed cRBEF expert remained slightly above the complete ReCoMER result in the formal test; EvidenceRouter was not beneficial on the tested M3ED representation; and the abstention variants did not improve every dataset. The evidence therefore does not support a claim of universal gains from every module. Instead, it indicates that history compensation, contribution routing and expert fusion are most useful when the corresponding context or modality evidence is reliable, and that their effects depend on the feature regime and evaluation protocol.

Overall, ReCoMER provides a reproducible and interpretable way to combine historical context, modality-specific evidence and an independent expert in multimodal conversational prediction. Its main value is the explicit separation of prediction, history admission, contribution estimation and expert fusion, together with diagnostics that expose when these signals help or fail. Future work should evaluate all variants under a single unified feature protocol, complete same-protocol external baselines and subgroup analyses, and calibrate history rejection and expert weighting for more stable cross-dataset transfer.

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